UIR الجامعة الدولية للرباط
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Université Internationale de Rabat
L'UNIVERSITÉ INNOVANTE

- TP 5 : Serverless Functions
- Professor: Ouassim Karrakchou
- Level : 4th year

TP 5 – Part 2

Serverless Functions

Objectives:

The goal of this TP is to introduce you to Serverless Functions in Microsoft Azure.

We will do the following tasks:

- Deploy Serverless Functions in Azure
- Secure calls to the Serverless Function
- Monitor the Serverless Function in Azure

Evaluation:

These TPs have two parts that are evaluated:

1. Experimental part: call the Professor to verify what you did when it tells you to do so.
2. Assignment: an assignment needs to be done at the end of the class. Please send it by email to your TP teacher **before midnight the day of the TP**:
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    b. Mohammed SOUIDI souididoc@gmail.com
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The TP must be done in groups of 2 to 3 students. The groups must stay the same for all the next TPs.

I- Pre-requisites:

In this TP, we will deploy the Serverless function we developed in the part 1 of this TP in Azure. You need thus to have a Serverless function that works locally. If you haven't finished TP5 part 1, make sure you at least have done the TP5 part 1 up to section 3 (to have the function working locally).

II- Deploy the function to Azure:

1. Open the folder containing the code of your Serverless function in Visual Studio Code.
2. Allow the **GET** method in your function if you removed it in TP5 part 1.
3. Click on the **Azure** icon in Visual Studio Code.
4. Right-click on **Function App** under **Ressources** and click on **"Create new Function App in Azure…"**. Enter the following parameters:
    1. Name: enter a name for your function app, it needs to be unique in all of azure, as it will be part of the url of your serverless function.
    2. Runtime Stack: select Python 3.14
    3. Location: select the Azure region you usually use
    4. Select Secrets
5. A new **resource group** will be created, along with several services needed to run your function app. Connect to your Azure portal: what resources were created?
6. Go back again to the **Azure** menu in Visual Studio Code. Select your function project in the **Workspace** section and click on the little cloud icon ☁ to deploy it in Azure. (There is sometimes a bug where the function is not deployed. You just have to do this step again).
7. Select the function app resource you just created to start the deployment.
8. Call the professor to show him.

III- Test the function deployed in Azure:

1. Go to the Azure portal and click on the Function App function resource.
2. Get the URL of the app function and try to access it in your browser. What do you see?
3. To get to the function, you have to complete your url by adding the following string at its end: **/api/addfunction**

4. Try now to access the full url of the function in your browser. What happens? What HTTP error do you get, and what does it mean?
5. Call the professor to show him.
6. Look at the following web page https://learn.microsoft.com/en-us/azure/azure-functions/functions-bindings-http-webhook-trigger at the **Authorization Level** and **Function Access Keys** sections. Remember that the authorization level you chose in TP5 part 1 was **Function**. Based on what is said in the web page, what do you need to do?
7. Retrieve the default function key by clicking first on the name of your function (**AddFunction**) in the bottom of the **Overview** page of your **function app** in the Azure Portal, then clicking on **Function keys**. Add this function key to the URL of your function as shown in the documentation (https://learn.microsoft.com/en-us/azure/azure-functions/functions-bindings-http-webhook-trigger).
8. Try now to access the full url of the function with the key in your browser. Does it work?
9. Call the professor to show him.

IV- Monitor the function deployed in Azure:

1. Go to the Azure portal and click on the Azure application insights resource.
2. What do you see? Explain what this resource is used for.
3. Call the professor to show him.

V- Call your function app from a Python script:

1. Update the script you used in TP2 to query the image classifier to now send a POST request to your function deployed in Azure. Pass any parameters needed to your function.
2. Call the professor to show him.

VI- Call your function app from an HTML page:

1. Create a simple HTML page that will send a **POST** request to your function deployed in Azure using the **Fetch** Javascript API and display its result in the page. Pass any parameters needed to your function. (Hint: if you have an issue, make sure to use the development tools of Google Chrome to check what happens).
2. Call the professor to show him.

3. You'll have to update the CORS policy of the function to allow requests from all hosts. Search how it is done in the Azure portal and do it to solve the issue.
4. Call the professor to show him.

VII- Assignment:

Based on what you learned in the TP explain the different steps needed to deploy an HTTP trigger function to the cloud, secure it, monitor its execution in the cloud, and call it from a web page or a Python script. Make sure to explain the different steps you do and use screenshots. Also, make sure you add your answers to the questions in sections III and VI, as well as the code of the script and the web page. Don't forget at the end to delete the resources in Azure.